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AIR AND UNDERWATER HEARING CHARACTERISTICS OF THE NORTHERN FUR SEAL: AUDIOGRAMS, FREQUENCY AND DIFFERENTIAL THRESHOLDS*

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Using the technique of instrumental conditioned reflexes the authors have investigated the underwater (in the frequency range 05–40 kHz) and air (in the range 0·1–25 kHz) audiograms of the northern fur seal and also the sensitivity of its hearing to change in the frequency of sound in the two media. The lowest thresholds of detection of tonal pulses (duration 100 msec, repetition frequency 5 pulse/sec) were recorded at the frequencies 5–17 kHz in water (–41 . . –45 dB relative to 1 μbar) and 2–16 kHz in air (11–20 dB relative to 0·0002 dyne/cm^2). It was established that the sensitivity of hearing of the fur seal to underwater sounds is 15–20 dB (in the main frequency range of auditory perception) better than to air sounds; the minimum frequency differential thresholds (f.d.t.) on discrimination of pure and frequency-modulated tones amount to 1·6–1·8% (3–5 kHz) in air and 1–2% (1–30 kHz) in water. It is shown that both in water and in air the regions of sharp rise in the f.d.t. coincide with the upper boundary frequencies of the auditory perception ranges.

In a series of investigations of marine animals, far from the last place is occupied by seals freely orienting in two media largely through the auditory analyser.

It has been demonstrated experimentally that pinnipedia hear in water no worse than humans in air (when compared at frequencies of optimum sensitivity). The auditory thresholds of seals in an air medium exceed by 10–35 dB (depending on the species and the frequency of sound) those in water and vary little within a family differing in the main in the representatives of different families [1–14]. The range of underwater sounds differentiated by frequency is confined for true seals to the frequency 60 kHz [13, 15, 16] and for the sea lion to 32–36 kHz [10, 11]. Analysis of audiograms showed that both true seals and sea lions possess two modes of sound conduction under water: (1) direct, tympanal; and (2) due to bone conductivity and not allowing ultrasounds to be discriminated by frequency [10]. Evidently, sharp rise in the differential frequency thresholds may serve as a criterion for determining the boundaries of the frequency range of auditory perception.

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In the work we measured the underwater and air audiograms and also the frequency differential thresholds (f.d.t.) in two media in the northern fur seal (*Callorhinus ursinus*) — adult 8–9-year-old male.

METHODS

The experiments were conducted in a concrete basin measuring 27·5 × 9·5 × 4·5 m situated in a room measuring 80 × 15 × 8 m, using the technique of instrumental conditioned reflexes. At the start the seal was in one of two positions: either in the horizontal (under the water near the surface) for measurement in an aqueous medium, or in the vertical (head above the water) for measurement in an air medium, the nose touching the starting manipulator.

To investigate the underwater audiogram the emitter (cylindrical magnetostriction transducer 120 mm high, diameter 35 mm) was suspended at a depth of 1 m and a distance of 2·6 m from the starting point of the animal. For the investigation of the air audiogram the emitter (AS-416 or MAS-10 dynamic 25) was positioned at a height of 0·5 m relative to the water surface at a distance of 3·8 m from the starting point.

In expectation of a signal the seal remained in the starting position but on switching on the sound the seal approached its source. The level of the signal discretely (with a step 3–6 dB) fell to the threshold value corresponding to the 75% level of positive reactions. To determine the detection thresholds the duration of the tonal pulses was 100 msec, the repetition frequency 5 pulse/sec and the duration of presentation of the signals 1 sec. The sensitivity of hearing of the fur seal to change in the frequency of sound was determined by a similar technique with the difference that on switching on a pure sound (positive signal) the seal approached the sound source and with the arrival of a frequency modulated (negative) tonal signal it remained at the starting manipulator. The f.d.t. was determined from the frequency deviation corresponding to the 75% level of positive reactions.

RESULTS

In Fig. 1 the underwater audiogram of the northern fur seal is given as compared with the frequency-threshold characteristics of hearing of the common seal and the Californian sea lion [10]. In the range of frequencies studied, 0·5–40 kHz, the region of highest sensitivity of hearing of the fur seal in an aqueous medium comes at the frequencies 5–17 kHz (–41 . . –45 dB relative to 1 μbar). The underwater audiogram of our test animal was close to the audiograms obtained for this species by other authors [7, 9].

From Fig. 2 one may compare the air frequency-threshold characteristic of our test fur seal with the corresponding curves of the audibility of the Californian sea lion, the common seal and humans.

The sounds presented in air are perceived by the fur seal in the range of frequencies studied 0·1–25 kHz with the highest sensitivity at the frequencies 2–16 kHz (11–20 dB relative to 0·0002 dyne/cm^2). As in water, the possibilities of detection by the fur seal of air sounds are closer to those in true seals. It is significant that in both aquatic and air media the differences in the sensitivity of hearing of the fur seal and other pinnipeds (in the main frequency range of auditory

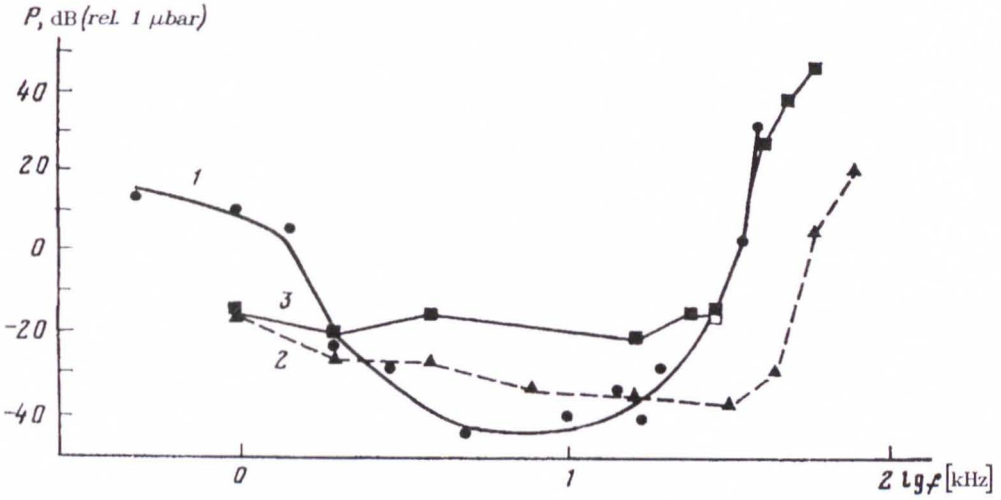


Fig. 1. Underwater audiograms of: *1*, the northern fur seal; *2*, the common seal [5]; *3*, the Californian sea lion [10].

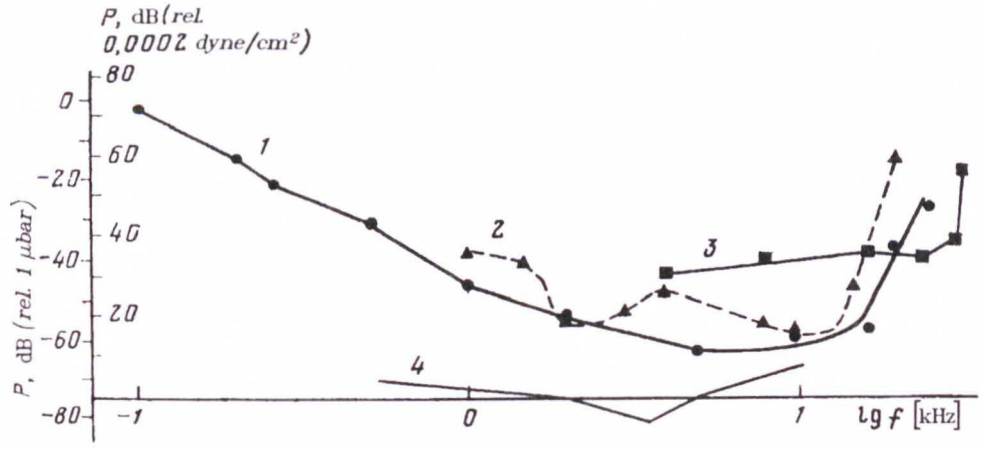


Fig. 2. Air audiograms of: *1*, the northern fur seal; *2*, the common seal [5]; *3*, the Californian sea lion [8]; *4*, humans [17].

perception) remain practically unchanged: the common seal is inferior to the fur seal, on average, by 10 dB (and this only at frequencies close to 5 kHz) and the sea lion by 20–25 dB.

Figure 3 presents the underwater and air audiograms of the northern fur seal in intensity–frequency coordinates, allowing one to evaluate correctly the difference in the sensitivity of hearing in the two media differing in acoustic resistance. As may be seen, the sensitivity of hearing of the northern fur seal in water is 15–20 dB (in the main frequency range of auditory perception) better than in air.

The frequency differential thresholds were investigated in the same fur seal over the frequency range 0·5–70 kHz in the aquatic and 0·1–25 kHz in the air medium at the modulation frequency 3 Hz and for a sinusoidal law of frequency modulation. The level of sensation of the signals was 20–60 dB. The minimal f.d.t. of the fur seal come at the frequencies 3–5 kHz in air (1·6–1·8%) and 1–30 kHz in water (1–2%) (Fig. 4). Outside these frequency zones the f.d.t. rise steeply, reaching 14% at 40 kHz in water and over 16% at 25 kHz in air.

With increase in the modulation frequency by 1–6 Hz the f.d.t. values (in an air medium at the frequency 10 kHz) fell from 9·4 to 0·48%. Change in the law of frequency modulation from sinusoidal to rectangular was accompanied by a slight fall in the thresholds — by 0·5 and 1·4%

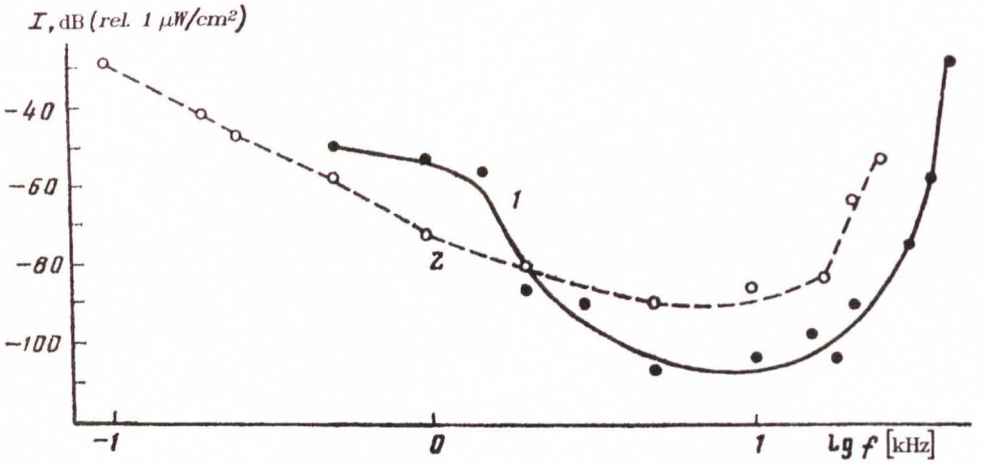


Fig. 3. Sensitivity of hearing of the northern fur seal in water (*1*) and air (*2*).

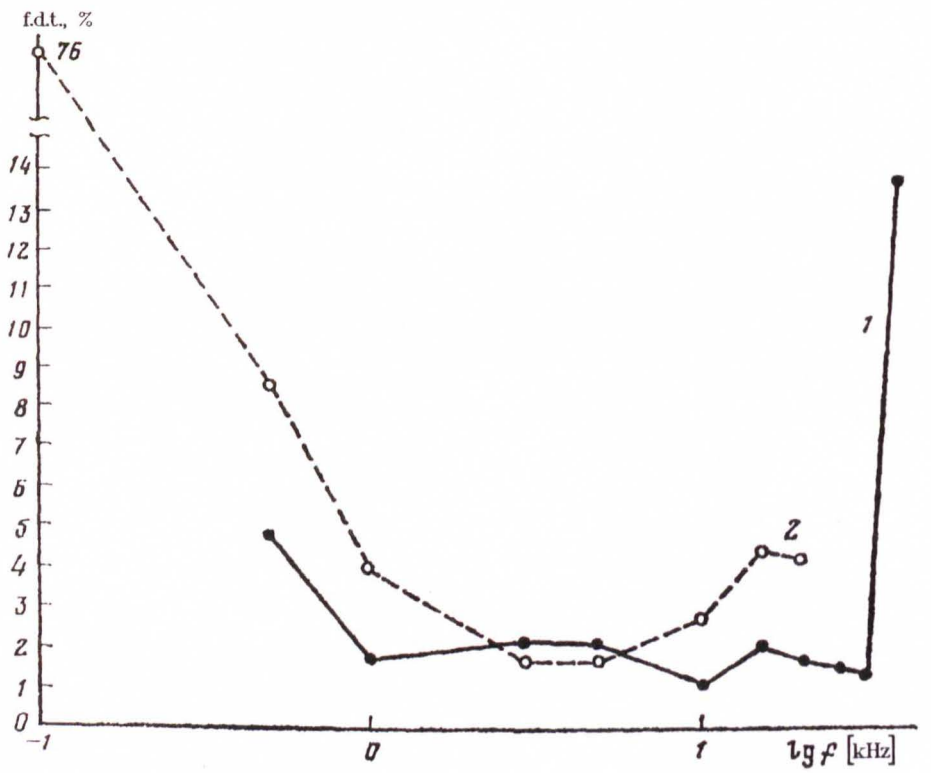


Fig. 4. Sensitivity of hearing of the northern fur seal to change in frequency in aquatic (*1*) and air (*2*) media.

at the frequencies 10 and 1 kHz, respectively. If it is assumed that the sharp rise in the f.d.t. is due to passage beyond the frequency range of auditory perception then its upper boundary in the northern fur seal lies at frequencies between 20 and 25 kHz for the air medium and 30 and 40 kHz for the aquatic medium, in line with the measured audiograms.

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